

Umit Y. Basaran

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Education

Ph.D. in Electrical Engineering, University of California, Riverside, CA 2022 – Present
Dean's Distinguished Fellowship Award | Advisor: Prof. Basak Guler

B.Sc. in Computer Science, Bilkent University, Turkey 2017 – 2022
Full-tuition merit scholarship (top 0.01% nationwide)

Research Interests

Certified Machine Unlearning • Privacy-preserving ML • Federated Learning • Secure MPC • Differential Privacy

Publications

- **U. Y. Basaran**, S. M. Ahmed, A. K. Roy-Chowdhury, and B. Guler, “A certified unlearning approach without access to source data,” *International Conference on Machine Learning (ICML)*, 2025 (accepted).
- **U. Y. Basaran**, S. M. Ahmed, D. S. Raychaudhuri, A. Dutta, R. Kundu, F. F. Niloy, B. Guler, and A. K. Roy-Chowdhury, “Towards source-free machine unlearning,” *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 4948–4957, 2025 (accepted).
- X. Lu, **U. Y. Basaran**, and B. Guler, “Scalable multi-round multi-party privacy-preserving neural network training,” *IEEE Transactions on Information Theory*, vol. 70, no. 11, pp. 8204–8236, 2024.
- **U. Y. Basaran**, X. Lu, and B. Guler, “Beyond secure aggregation: Scalable multi-round secure collaborative learning,” *Federated Learning and Analytics in Practice*, 2023.

Research & Engineering Experience

Graduate Research Assistant, University of California, Riverside, CA Sep 2022 – Present
Advisor: Prof. Basak Guler

- Developed **certified machine unlearning** algorithms that remove the influence of deleted data **without access to source samples**, using noise calibration and linearization; evaluated on CIFAR-10/100 with guarantees on forget scores, accuracy, and membership-inference robustness (**ICML 2025, CVPR 2025**).
- Studied **information-limited unlearning** setups (e.g. lack of data, manipulated models, etc.), comparing against retrain-from-scratch and existing approximate unlearning baselines and characterizing accuracy-vs-unlearning tradeoffs.
- Designed **multi-round privacy-preserving training protocols** over finite fields for multi-party settings, analyzing communication and computation overheads and proving security guarantees; work led to publications in **IEEE TIT 2024** and **FLAP 2023**.

Research Intern, HyperbeeAI, San Francisco, CA Mar 2021 – Jun 2022

- Investigated **neural network compression** for edge deployment, combining quantization, pruning, binarization, and knowledge distillation for ResNet-style architectures; implemented compressed PyTorch models that reduced memory and compute budgets while maintaining competitive accuracy on internal benchmarks.

ML Research Intern, ASELSAN (Turkey's largest defense contractor), Ankara, Turkey Sep 2021 – Dec 2021

- Explored **domain adaptation** for day/night aircraft detection in safety-critical radar systems, benchmarking adversarial and feature-alignment methods on an internal ~50K-image dataset and improving night-time detection accuracy from roughly 60–65% to 80–90%.

Research Intern, Karel Electronics, Ankara, Turkey Sep 2020 – Mar 2021

- Designed ML-based **patient monitoring prototypes** for individuals with multiple sclerosis and Parkinson's disease using triaxial accelerometer streams, building lightweight models that achieved around 90% detection accuracy in a year-long pilot with 5–6 patients.

Technical Skills

Programming: Python, C, C++

ML Frameworks: PyTorch, TensorFlow, JAX

ML Topics: Privacy-preserving machine learning, certified machine unlearning, differential privacy, multi-party computing, model compression, domain adaptation, information theory

Tools: L^AT_EX, Git, Docker, Linux, Obsidian, Zotero

Academic Service & Awards

Reviewer: IEEE SSP Workshop (2025); IEEE Transactions on Signal Processing (2024)

Awards: Dean's Distinguished Fellowship (UCR); Full-tuition Scholarship (Bilkent University)